## Problem Summary

Given the head of a singly linked list :

$$L_0 \rightarrow L_1 \rightarrow L_2 \rightarrow \dots \rightarrow L_n$$

Reorder it to

* Must modify in-place

$$L_0 \rightarrow L_n \rightarrow L_1 \rightarrow L_{n-1} \rightarrow L_2 \rightarrow L_{n-2} \rightarrow \dots$$

## Key Idea

This problem has 3 parts:

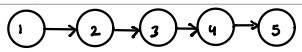
1) Find the middle of the list

2) Reverse the second half of the list

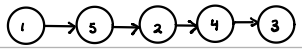
3) Merge the two halves alternatingly

## Diagram Example

Before:




After:




## Approach

1) If the list is empty, return

2) Find middle using slow and fast

3) Reverse from middle onward

4) Set first = head, second = slow

5) Merge the two lists

6) Continue until second half is used up

Time Complexity: $O(n) \rightarrow$ Overall since finding middle is $O(n)$, reverse half is $O(n)$, and merging is $O(n)$

Space Complexity: $O(1) \rightarrow$ in-place reordering